

Document Summary:

(1). Software used; (2). References and software operation videos; (3). Specific operation settings files and data selections.

(1) Software used: Notepad++, Simens NX & Spaceclaim, Fluent Mesh, PSPad editor , Tecplot.

(2) References and software operation: Reference 0-17

Reference 0. Overset mesh

<https://zhuanlan.zhihu.com/p/574164035>

Reference 1. Fluent-2D Dynamic Mesh Settings

https://www.bilibili.com/video/BV1454y1J7sx/?spm_id_from=333.788.video.desc.click&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 2. Overlapping Grid Settings

https://www.bilibili.com/video/BV1234y1X7kj/?spm_id_from=333.788.video.desc.click&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 3. Pitch motion – using overlapping and moving meshes to simulate dynamic stall.

https://www.bilibili.com/video/BV1ru411i71G/?spm_id_from=333.788.recommend_more_video.12&trackid=web_related_0.router-related-2206146-52fdn.1761546209006.892&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 4. [Fluent's Comprehensive Tutorial] Everything I Know About 2D Flapping Wing Simulation is Here

https://www.bilibili.com/video/BV1f64y1b7Hb/?spm_id_from=333.337.search-card.all.click&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 5. Fluent Case 33: Flexible Flapping Wing - Gas-Solid Two-Way Coupling

https://www.bilibili.com/video/BV1XY411g7vz/?spm_id_from=333.337.search-card.all.click&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 6. simulation of flapping wing in ansys fluent

<https://www.youtube.com/watch?v=gozPMCE4kfE>

Reference 7. [Fluent Case Study] Fish Swimming in Water UDF: Dynamic Mesh + Overlapping Mesh

https://www.bilibili.com/video/BV1cL411A7aW?spm_id_from=333.788.recommend_more_video.12&trackid=web_related_0.router-related-2206146-wxmlq.1761554102149.600&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 8. Two-dimensional airfoil mesh independence and boundary condition settings

https://www.bilibili.com/video/BV12X4y167Qu?spm_id_from=333.788.recommend_more_video.13&trackid=web_related_0.router-related-2206146-wxmlq.1761634099613.597&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 9. 3D wing modeling, simulation, and post-processing

https://www.bilibili.com/video/BV18c411t7Bn?spm_id_from=333.788.recommend_more_video.1&trackid=web_related_0.router-related-2206146-n6d7k.1761634120942.710&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 10. Fluent Case Study 108 NACA0012: Three-Dimensional Flow in an Airfoil

https://www.bilibili.com/video/BV1Bs4y1U7jE?spm_id_from=333.788.recommend_more_video.0&trackid=web_related_0.router-related-2206146-n7lrp.1761634540716.468&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 11. Ansys-Fluent-NACA0018 Airfoil Numerical Simulation (3D Modeling + Meshing + Simulation Analysis)

https://www.bilibili.com/video/BV11g411k7eB?spm_id_from=333.788.recommend_more_video.3&trackid=web_related_0.router-related-2206146-5l4x4.1761634641240.868&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 12. Overset Mesh Generation Based on Ansys Fluent – 1/2

https://www.bilibili.com/video/BV1Ur4y1F7b7/?uid=425631557234793146376237&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 13. Overset Mesh Generation Based on Ansys Fluent – 2/2

https://www.bilibili.com/video/BV1rr4y1F73z/?uid=42563172723479314637337A&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 14. Fluent Case Study 103: NACA Airfoil Analysis - Airfoil Database Application

https://www.bilibili.com/video/BV1hM4y167dd/?spm_id_from=333.337.search-card.all.click&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 15. For project details, refer to the YouTube video: Fluid Structure Interaction Analysis on Aircraft Wing | Ansys CFX | Pressure Mapping | 1-way FSI

<https://www.bing.com/videos/riverview/relatedvideo?q=Fluid+Structure+Interaction+Analysis+on+Aircraft+Wing+%7c+Ansys+CFX+%7c+Pressure+Mapping+%7c+1-way+FSI&qs=n&sp=-1&lq=0&pq=fluid+structure+interaction+analysis+on+aircraft+wing+%7c+ansys+cfx+%7c+pressure+mapping+%7c+1-way+fsi&sc=0-96&sk=&cvid=175C3B79A7C84CAFA84766C2BDA6CEEE&ajaxnorecss=1&sid=0682908F01816B6D319D861B00B16A5E&jsoncbid=0&ajaxsydconv=1&ru=%2fsearch%3fq%3dFluid%2520Structure%2520Interaction%2520Analysis%2520on%2520Aircraft%2520Wing%2520%257C%2520Ansys%2520CFX%2520%257C%2520Pressure%2520Mapping%2520%257C%25201-way%2520FSI%26qs%3dn%26form%3dQBRE%26sp%3d->

[1%26lq%3d0%26pq%3dfluid%2520structure%2520interaction%2520analysis%2520on%2520aircraft%2520wing%2520%257C%2520ansys%2520cfx%2520%257C%2520pressure%2520mapping%2520%257C%25201-way%2520fsi%26sc%3d0-96%26sk%3d%26cvid%3d175C3B79A7C84CAFA84766C2BDA6CEEE%26ajaxnorecss%3d1%26sid%3d0682908F01816B6D319D861B00B16A5E%26format%3dsnrjson%26jsoncbid%3d0%26ajaxsydconv%3d1&mmscn=vwrc&mid=C975AA4E3AA09F975D5BC975AA4E3AA09F975D5B&FORM=WRVORC&ntb=1&msockid=6edbc503b49d11f0af613e6449f72781](https://www.bilibili.com/video/BV1W34y1D7jP/?spm_id_from=333.1387.homepage.video_card.click&vd_source=2db8a90eb811a6b91ddaa382561fa186)

Reference 16. [FLUENT's Step-by-Step Tutorial] Negative Volume in Moving Meshes? Overlapping Meshes Solved in One Step! (Also Includes a Super Detailed Explanation of Tecplot Post-Processing)

https://www.bilibili.com/video/BV1W34y1D7jP/?spm_id_from=333.1387.homepage.video_card.click&vd_source=2db8a90eb811a6b91ddaa382561fa186

Reference 17. SpaceClaim import airfoil point curve file operation

https://www.bilibili.com/video/BV1ftgne6EWk/?vd_source=2db8a90eb811a6b91ddaa382561fa186

(Press the F key (Fit View) on your keyboard, and SpaceClaim will automatically center and display all geometry in the view.)

(3) Specific operation settings files and data selections: naca0012.txt, udf file, vortex.scm, 2D model(background&foremesh), Mesh(background&foremesh), Final Case, Animation Video(with_vector & without_vector).

[All files can be found at <https://github.com/Pika-Zhihao/2.Flapping-Wing-Simulation-Fluid-Structure-Interaction>]